Operator Precedence

Operator precedence determines the order in which the operators in an expression are evaluated.

For eg – *int x = 3 * 4 - 1;*

In the above example, the value of x will be 11, not 9. This happens because the precedence of * operator is higher than - operator. That is why the expression is evaluated as (3 * 4) - 1 and not 3 * (4 - 1).

Operator Precedence Table

Operators	Precedence
postfix increment and decrement	++ --
prefix increment and decrement, and unary	++ -- + - ~ !
multiplicative	* / %
additive	
shift	<< >> >>>
relational	< > <= >= instanceof
equality	== !=
bitwise AND	&
bitwise exclusive OR	^
bitwise inclusive OR	|
logical AND	&&
logical OR	||
ternary	? :
assignment	= += -= *= /= %= &= ^= |=
	<<= >>= >>>=

Associativity of Operators

If an expression has two operators with similar precedence, the expression is evaluated according to its associativity (either left to right, or right to left).

Operators	Precedence	Associativity
postfix increment and decrement	++ --	left to right
prefix increment and decrement, and unary	++ -- + - ~ !	right to left
multiplicative	* / %	left to right
additive	+ -	left to right
shift	<< >> >>>	left to right
relational	< > <= >= instanceof	left to right
equality	== ! =	left to right
bitwise AND	&	left to right
bitwise exclusive OR	^	left to right
bitwise inclusive OR		left to right
logical AND	&&	left to right
logical OR	||	left to right
ternary	? :	right to left
assignment	= += -= *= /= %= &= ^= | = <<= >>= >>>=	right to left

Note - These notes are just for a quick glance. We don't have to memorize them all at once. Most of these rules are very logical and we have been following them in a lot of instances already.